

TUPO MICROFINANCE

PBX SYSTEM CONFIGURATION DOCUMENTATION

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INTRODUCTION

This document provides a formal record of the implementation and configuration of the IP-based PBX and Gateway system for TUPO Microfinance. The system was deployed to centralize communication, improve customer call handling, and enhance operational efficiency.

Scope of the Project

The scope of this project included:

- Deployment of a Yeastar S20 IP PBX system
- Integration of Yeastar TG800 GSM Gateway
- Configuration of internal SIP extensions
- Implementation of IVR and inbound call routing
- Configuration of outbound call rules
- System testing and validation

System devices

1. Yeastar S20 IPPBX (1)
2. Yeastar TG800 GSM (1)
3. NETGEAR 16-port Gigabit Ethernet PoE+ Unmanaged switch (1)
4. FANVIL X301P IP phones (8)

SYSTEM CONFIGURATION

NETWORK CONFIGURATION:

The following static Ip addresses were assigned to the devices.

Yeastar S20 IPPBX - **192.168.100.254**
Yeastar TG800 GSM - **192.168.100.253**
Gateway - **192.168.100.1**

Admin portal

IPPBX:

Yeastar S20 - **192.168.100.254**
Username - **admin**
Password - **Tupo@@2026**

TG800 GSM

Yeastar TG800 -**192.168.100.253**
Username -**admin**
Password -**Tupo@@2026**

GSM GATEWAY CONFIGURATION

The TG800 GSM Gateway was configured with eight (8) SIM cards to handle customer calls.

Mobile list

8 GSM ports are named as LINE 1-LINE8

Yeastar						
▼ Mobile List	Mobile List					
Mobile List	Port	Trunk Name	Type	Talk Time(min)	Used(min)	Available(min)
Mobile Group	1	LINE1	Mobile	Unlimited	0	Unlimited
▼ VoIP Settings	2	LINE2	Mobile	Unlimited	0	Unlimited
VoIP Trunk	3	LINE3	Mobile	Unlimited	3	Unlimited
Trunk Group	4	LINE4	Mobile	Unlimited	0	Unlimited
SIP Settings	5	LINE5	Mobile	Unlimited	0	Unlimited
IAX Settings	6	LINE6	Mobile	Unlimited	0	Unlimited
General Preferences	7	LINE7	Mobile	Unlimited	0	Unlimited
	8	LINE8	Mobile	Unlimited	0	Unlimited

Mobile group

Two mobile groups were created

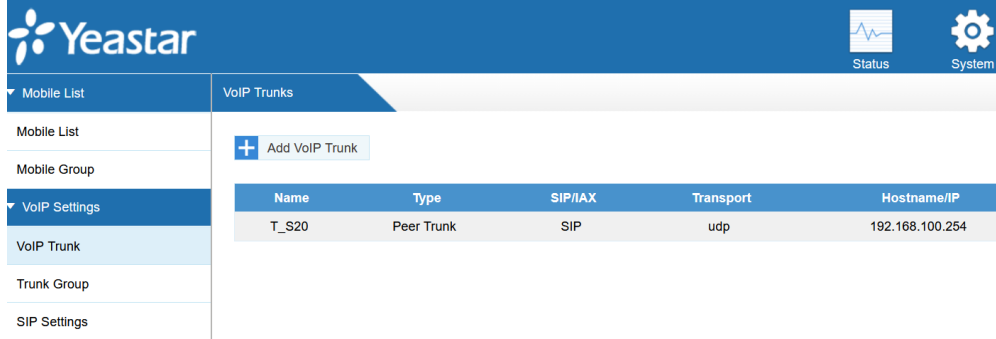
- Call_center -this includes LINE1 and LINE2 which will be main for incoming calls.
- TUPOM -this includes the remaining LINE3 To LINE8

Yeastar			Status
▼ Mobile List	Mobile Group		
Mobile List			
Mobile Group	+ Add New Mobile Group ⓘ		
▼ VoIP Settings			
VoIP Trunk			
Trunk Group			
SIP Settings			
IAX Settings			

Group Name	Strategy	Group Members
TUPOM	Balance	LINE3-LINE4-LINE5-LINE6-LINE7-LINE8
Call_Center	Balance	LINE1-LINE2

Voip Trunk

Single Voip Trunk is created to create a connectivity between external phone number to the PBX



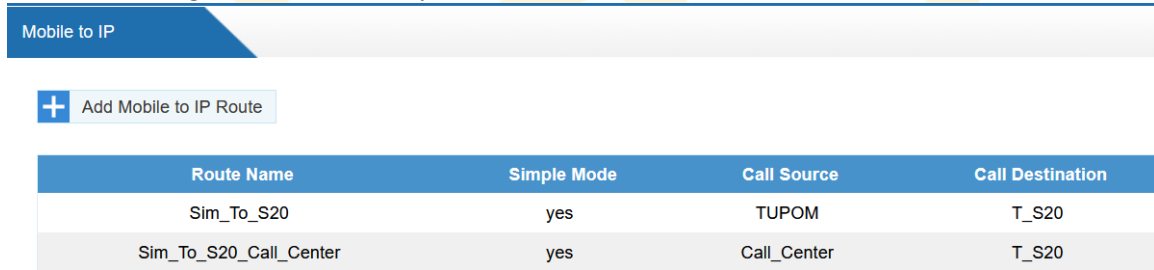
The screenshot shows the Yeastar web interface for VoIP Trunks. The left sidebar has a menu with 'Mobile List', 'Mobile Group', 'VoIP Settings', 'VoIP Trunk', 'Trunk Group', and 'SIP Settings'. The 'VoIP Settings' section is expanded, and 'VoIP Trunk' is selected. The main area shows a table with one entry, 'T_S20', which is a Peer Trunk using SIP transport over UDP, with a hostname/IP of 192.168.100.254. There is an 'Add VoIP Trunk' button above the table.

Name	Type	SIP/IAX	Transport	Hostname/IP
T_S20	Peer Trunk	SIP	udp	192.168.100.254

Mobile to IP

Mobile-to-IP routing has been configured to direct incoming calls from GSM SIM lines into the Yeastar S20 IP PBX.

Two routes were created to handle calls from general TUPO mobile lines and Call Center SIMs, both forwarding them to the S20 system.



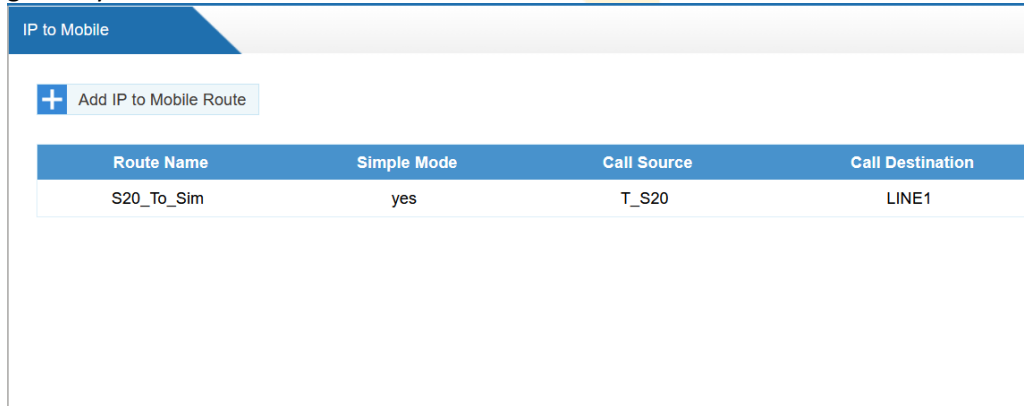
The screenshot shows the Yeastar web interface for Mobile to IP routing. The left sidebar has a menu with 'Mobile List', 'Mobile Group', 'VoIP Settings', 'VoIP Trunk', 'Trunk Group', and 'SIP Settings'. The 'VoIP Settings' section is expanded, and 'Mobile to IP' is selected. The main area shows a table with two entries, 'Sim_To_S20' and 'Sim_To_S20_Call_Center', both using Simple Mode. The 'Sim_To_S20' route has a call source of TUPOM and a call destination of T_S20. The 'Sim_To_S20_Call_Center' route has a call source of Call_Center and a call destination of T_S20. There is an 'Add Mobile to IP Route' button above the table.

Route Name	Simple Mode	Call Source	Call Destination
Sim_To_S20	yes	TUPOM	T_S20
Sim_To_S20_Call_Center	yes	Call_Center	T_S20

IP to Mobile

IP-to-Mobile routing has been configured to allow calls from the Yeastar S20 IP PBX to be sent out through a GSM SIM line.

A route was created to forward calls originating from the S20 system to **LINE1** on the GSM gateway.



The screenshot shows the Yeastar web interface for IP to Mobile routing. The left sidebar has a menu with 'Mobile List', 'Mobile Group', 'VoIP Settings', 'VoIP Trunk', 'Trunk Group', and 'SIP Settings'. The 'VoIP Settings' section is expanded, and 'IP to Mobile' is selected. The main area shows a table with one entry, 'S20_To_Sim', using Simple Mode. The call source is T_S20 and the call destination is LINE1. There is an 'Add IP to Mobile Route' button above the table.

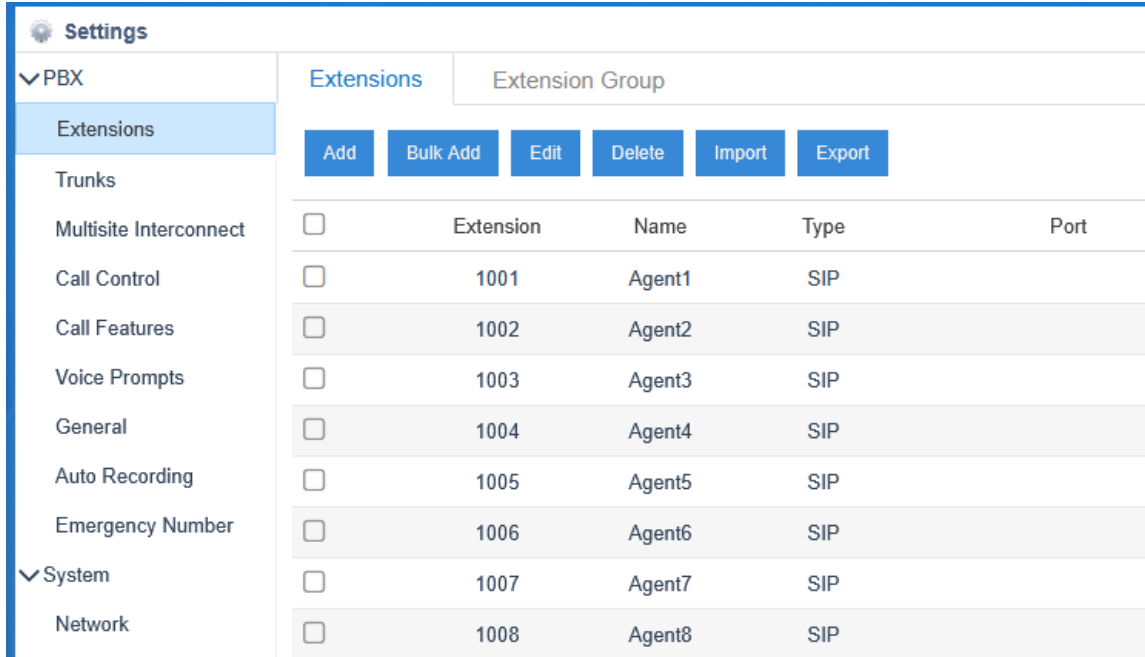
Route Name	Simple Mode	Call Source	Call Destination
S20_To_Sim	yes	T_S20	LINE1

IPPBX S20 CONFIGURATIONS

Extension Configuration

A total of 8 SIP extensions (1001–1008) have been configured on the PBX system. All incoming calls are first routed to SIM Extension 1 and SIM Extension 2 as the primary call-handling lines.

If these primary extensions are busy or unavailable, calls are automatically forwarded to the remaining extensions (1003–1008) as a fallback to ensure no call is missed.



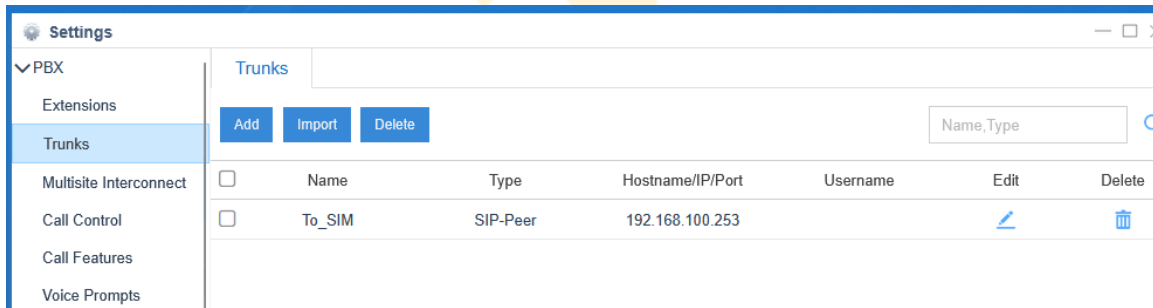
The screenshot shows the 'Settings' interface for a PBX system. The left sidebar is expanded to 'PBX' and then 'Extensions'. The main area has tabs for 'Extensions' and 'Extension Group'. Below the tabs are buttons for 'Add', 'Bulk Add', 'Edit', 'Delete', 'Import', and 'Export'. A table lists 8 SIP extensions, each with a checkbox, extension number, name, type, and port.

☐	Extension	Name	Type	Port
☐	1001	Agent1	SIP	
☐	1002	Agent2	SIP	
☐	1003	Agent3	SIP	
☐	1004	Agent4	SIP	
☐	1005	Agent5	SIP	
☐	1006	Agent6	SIP	
☐	1007	Agent7	SIP	
☐	1008	Agent8	SIP	

Trunks

A SIP trunk (To_SIM) has been configured as the main connection between the PBX and the GSM gateway.

This trunk handles all inbound and outbound calls between the PBX system and the SIM cards.



The screenshot shows the 'Settings' interface for a PBX system. The left sidebar is expanded to 'PBX' and then 'Trunks'. The main area has buttons for 'Add', 'Import', and 'Delete'. A search bar labeled 'Name, Type' is present. A table lists the configured trunks, including 'To_SIM'.

☐	Name	Type	Hostname/IP/Port	Username	Edit	Delete
☐	To_SIM	SIP-Peer	192.168.100.253			

Call control

Inbound Route

This is configured to receive all calls made from the gateway.

Settings

▼ PBX

Extensions

Trunks

Multisite Interconnect

Call Control

Call Features

<

Inbound Routes

Outbound Routes

Outbound Restriction

AutoCLIP Routes

Add

Import

Delete

Name,

☐	Name	DID Pattern	Caller ID Pattern	Edit	Delete
☐	Call_Center	2,5		✎	🗑

Outbound Route

This is configured to allow all the number prefixes to be able to make calls out from the PBX.

Settings

▼ PBX

Extensions

Trunks

Multisite Interconnect

Call Control

Call Features

<

Inbound Routes

Outbound Routes

Outbound Restriction

AutoCLIP Routes

Add

Import

Delete

Name,Dial Pa

☐	Name	Dial Pattern	Edit	Delete
☐	ROUTE_OUT	X.	✎	🗑 🔒

Voice prompt

A custom voice prompt is set to give a brief introduction about the company during making incoming calls.

Settings

▼ PBX

Extensions

Trunks

Multisite Interconnect

Call Control

Call Features

Voice Prompts

General

Prompt Preference

System Prompt

Music on Hold

Custom Prompts

Record New

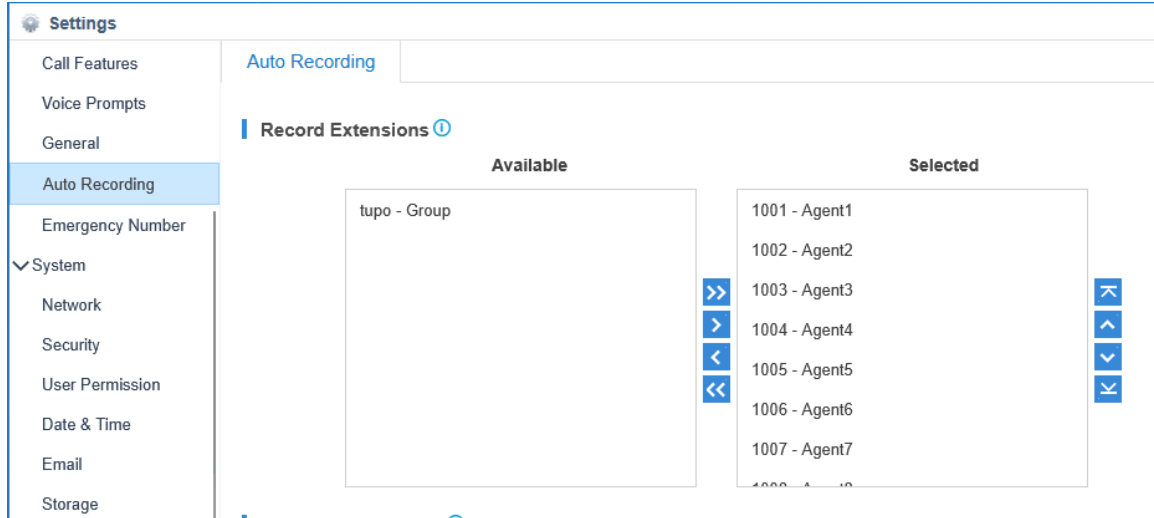
Upload

Delete

☐	Name	Record	Play	Download	Delete
☐	TUPO	🎤	▶	⬇	🗑

Auto Recording

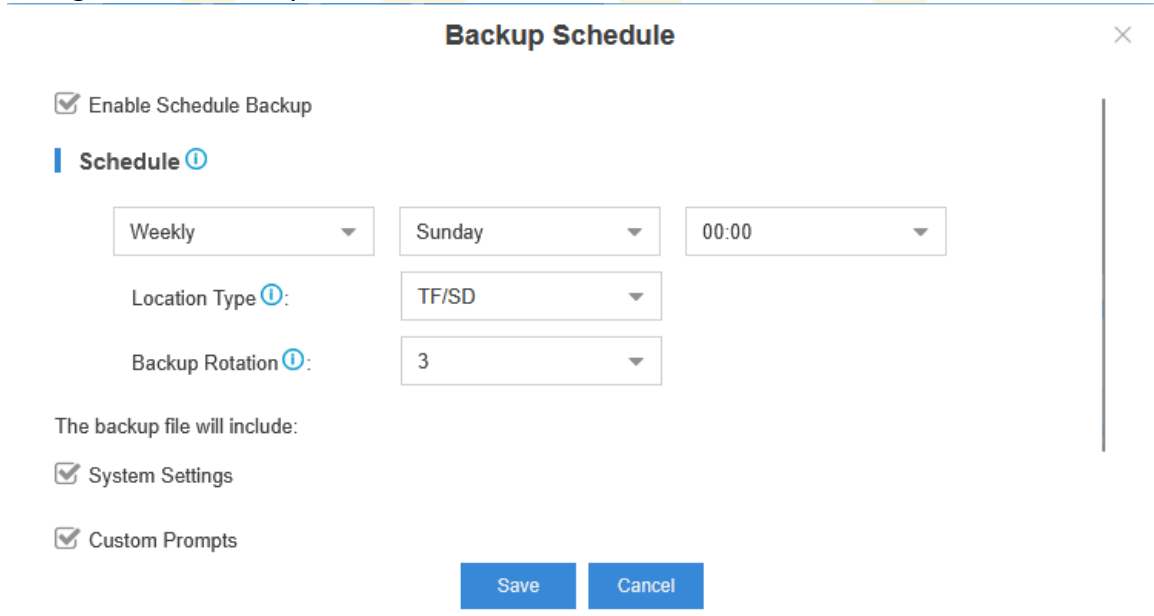
All incoming and outgoing calls from all extensions are recorded and stored in external storage.



The screenshot shows the 'Settings' menu on the left with 'Auto Recording' selected. The main area is titled 'Auto Recording' and contains a section 'Record Extensions' with a sub-header 'Record Extensions ①'. Below this, there are two columns: 'Available' and 'Selected'. The 'Available' column contains 'tupo - Group'. The 'Selected' column contains a list of extensions: '1001 - Agent1', '1002 - Agent2', '1003 - Agent3', '1004 - Agent4', '1005 - Agent5', '1006 - Agent6', and '1007 - Agent7'. There are blue arrow buttons between the columns for moving items back and forth.

Backup

Automatic weekly backups are enabled to run every Sunday at 00:00. Backups are stored on TF/SD storage with a rotation of 3 files to keep recent copies only. The backup includes system settings and custom voice prompts to ensure full configuration recovery.



The screenshot shows the 'Backup Schedule' window. At the top, there is a checkbox 'Enable Schedule Backup' which is checked. Below this is a section 'Schedule ①' with three dropdown menus: 'Weekly', 'Sunday', and '00:00'. Below these are two more dropdown menus: 'Location Type ①:' with 'TF/SD' selected, and 'Backup Rotation ①:' with '3' selected. Below these is a section 'The backup file will include:' with two checkboxes: 'System Settings' and 'Custom Prompts', both of which are checked. At the bottom, there are two buttons: 'Save' and 'Cancel'.

CONCLUSION

Post-implementation testing was carried out to ensure the PBX system, GSM gateway, and call flow logic are working reliably under real operating conditions. Multiple inbound and outbound call scenarios were simulated using different SIM lines and extensions to validate end-to-end functionality.

- **Inbound GSM calls** were successfully received through the configured SIP trunk from the GSM gateway, confirming proper trunk registration and signaling.
- **IVR navigation** was tested by dialing in and selecting menu options, with all key presses correctly detected and routed to the intended call paths.
- **Extension ringing behavior** was verified, ensuring calls rang on the correct primary extensions and followed the defined fallback sequence when extensions were busy or unavailable.
- **Outbound calls** from internal extensions to external GSM numbers were completed successfully, confirming correct outbound routing and gateway connectivity.
- **Call forwarding and failover logic** worked as designed, with calls automatically redirected to alternative extensions to prevent missed calls.

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